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END SEMESTER EXAMINATION - 2025

Name of the Course: **B. Tech Petroleum Engineering**

Semester: **Third**

Name of the Paper: **Advanced Mathematics**

Paper Code: **TMA-307**

Time: **3 Hours**

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b and c in each main question.
- (iii) Total marks in each main question are twenty.

Each question carries 10 marks.

Q1	(20 marks)						
(a)	Find the half-range sine series for $f(x) = x \cos x$ in $0 < x < \pi$						CO1
(b)	In the period of 4 obtain the Fourier series for $f(x) = x $, $-2 < x < 2$.						
(c)	Expand the function $f(x) = x$ as half-range cosine series in $0 \leq x \leq \pi$.						
Q2	(20 marks)						
(a)	Using Fourier integral representation, show that $\int_0^\infty \frac{\omega \sin \omega x}{1 + \omega^2} d\omega = \frac{\pi}{2} e^{-x}$, $x > 0$.						CO2
(b)	Show that $Z[n^p] = -z \frac{d}{dz} (Z[n^{p-1}])$, $n > 0$. Also find $Z[n]$ and $Z[n^2]$.						
(c)	Define DFT of a sequence. Also, find the DFT of the sequence $\{1, 2, 3, 4\}$.						
Q3	(20 marks)						
(a)	Illustrate the boundary value problem with example. Write the different types of partial differential equations which can be solved as boundary value problem.						CO3
(b)	A tightly stretched string with fixed ends $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is set vibrating by giving to each of its points an initial velocity $\lambda x(1 - x)$, find the displacement of the string at any distance x from one end at any time t .						
(c)	Find the temperature in a bar of length 2 whose ends are kept at zero and lateral surface insulated if the initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.						
Q4	(20 marks)						CO4
(a)	Find the mean and standard deviation for the following data:						
	Marks obtained	55-58	58-61	61-64	64-67	67-70	
	No. of students	12	17	23	18	11	

(b)	If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (a) Exactly 3 will suffer from a bad reaction. (b) None will suffer from a bad reaction																												
(c)	Find the mode and median of the data: <table><tr><td>Annual sales (in thousands)</td><td>0-5</td><td>5-10</td><td>10-15</td><td>15-20</td><td>20-25</td><td>25-30</td><td>30-35</td><td>35-40</td><td>40-45</td></tr><tr><td>Number of companies</td><td>29</td><td>195</td><td>241</td><td>117</td><td>52</td><td>10</td><td>6</td><td>3</td><td>2</td></tr></table>									Annual sales (in thousands)	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	Number of companies	29	195	241	117	52	10	6	3	2
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Number of companies	29	195	241	117	52	10	6	3	2																				
Q5 (20 marks)																													
(a)	Find a straight line to fit the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>3.3</td><td>4.5</td><td>6.3</td></tr></table>									x	0	1	2	3	4	y	1	1.8	3.3	4.5	6.3								
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(b)	A city reported that in an area 1705 men were literate and 1527 women were literate. Do the figures confirm to the hypothesis that the literacy of the men and women is in the ratio 1:2. Tabulated value at Z at 5% level of significance is 1.645.																												
(c)	The demand for a particular space part in a factory was found to vary from day-to-day. In sample study, the following information was found. <table><tr><td>Days</td><td>Mon</td><td>Tues</td><td>Wed</td><td>Thurs</td><td>Fri</td><td>Sat</td></tr><tr><td>No. of parts demanded</td><td>1124</td><td>1125</td><td>1110</td><td>1120</td><td>1125</td><td>1116</td></tr></table> Use Chi-square to test the hypothesis that the number of parts demanded is not dependent on the day of the week at 5% level of significance. Given for 5 df, χ^2 at the level of significance of 0.05 is 11.07									Days	Mon	Tues	Wed	Thurs	Fri	Sat	No. of parts demanded	1124	1125	1110	1120	1125	1116						
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CO5